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$$\begin{aligned}\frac{z}{z-2i} - \frac{3iz}{(z+i)(z-2i)} &= 2 \\ \Leftrightarrow \frac{z(z+i)}{(z-2i)(z+i)} - \frac{3iz}{(z+i)(z-2i)} &= 2 \\ \Leftrightarrow \frac{z^2 - 2iz}{(z-2i)(z+i)} &= 2 \\ \Leftrightarrow \frac{(z-2i)z}{(z-2i)(z+i)} &= 2 \\ \Leftrightarrow \frac{z}{z+i} &= 2 \\ \Leftrightarrow z &= 2z + 2i \\ \Leftrightarrow 0 &= z + 2i \\ \Leftrightarrow z &= -2i\end{aligned}$$

References